

# One-way ANOVA

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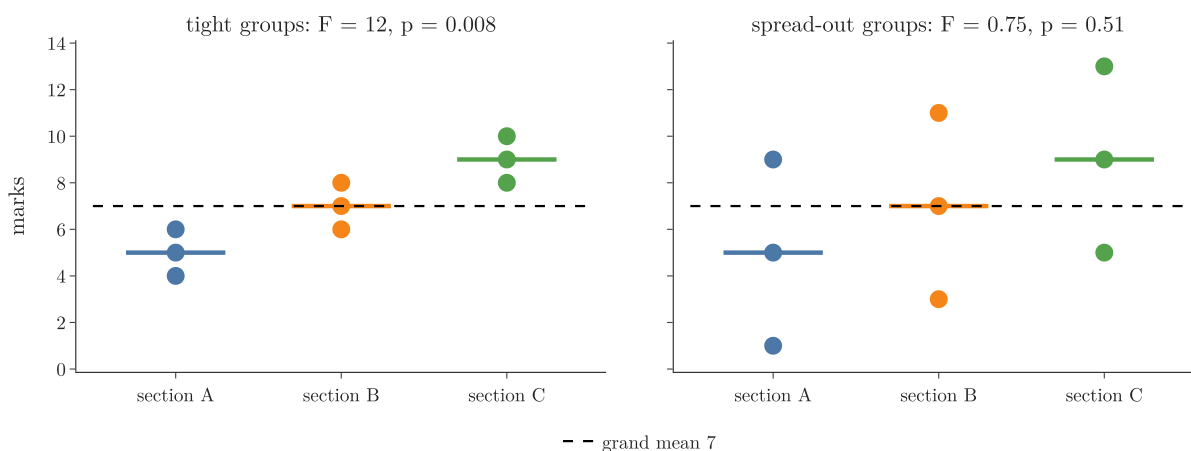
Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Variance ([Note ML-018](#)); Hypothesis testing: null and alternative ([Note MA-038](#)).
- **Compare with:** T-tests: one-sample, two-sample, paired ([Note MA-044](#)).

## 1. Overview

**Key point:** One-way ANOVA tests whether three or more group means are equal by comparing two variances: how far the group means lie from each other, and how far the values lie from their own group mean.



**Figure 1:** Same group means, different spread: the gaps between means are clear on the left and lost in the noise on the right

ANOVA (**analysis of variance** (G-203)) was named in [the tools of inferential statistics](#) as the test that compares the means of several groups. In [choosing a hypothesis test](#) ANOVA is the test for one numerical **feature** (G-772) against a categorical feature with three or more categories (a feature is a variable of the data, one column of the data table, such as age or class).

Figure 1 shows the whole idea. Both panels have the same group means, 5, 7 and 9. On the left the values sit close to their group mean, so the gaps between the means stand out. On the right the values are spread widely, and the same gaps could easily be chance. ANOVA turns this comparison into one number, the F statistic.

This Note covers:

- why we do not run a t-test for every pair of groups;

- the split of the total variation into within-group and between-group parts, worked by hand on nine marks;
- the hypotheses;
- the F statistic, the F distribution and the decision;
- the assumptions, a Titanic case study, and the follow-up question “which groups differ?”.

**Extra:** All of this Note is extra material beyond the “which test when” guide; the guide only names ANOVA.

## 2. Why not several t-tests?

**Key point:** Each t-test at  $\alpha = 0.05$  has a 5% false-alarm risk; three of them together have about a 14% risk, ten of them about 40%.

A two-sample t-test compares two group means (see [the independent two-sample t-test](#)). With three groups A, B and C, we could run three t-tests: A against B, A against C, B against C.

The trouble is the Type I error (see [Type I error](#)). Each test, run at  $\alpha = 0.05$ , wrongly rejects a true  $H_0$  5% of the time. Over several tests these risks add up.

1. **In words:** the chance of at least one false alarm is one minus the chance that every test stays quiet.
2. **Formula:** for  $m$  independent tests,

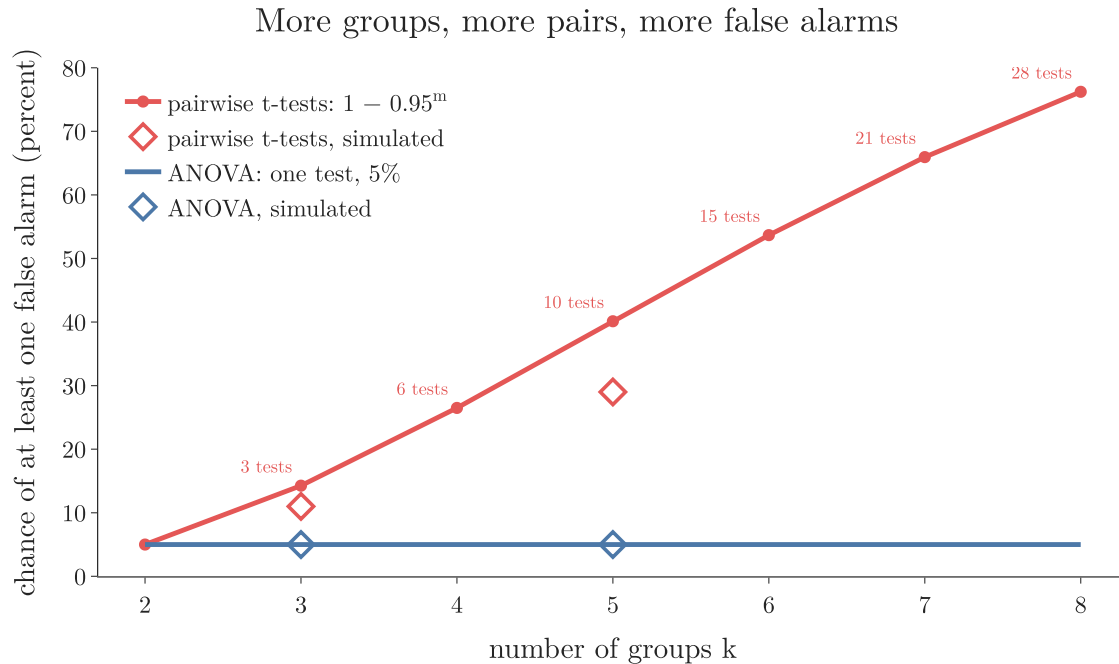
$$P(\text{at least one Type I error}) = 1 - (1 - \alpha)^m$$

3. **Example:** three groups need  $m = 3$  tests:

$$\begin{aligned} &1 - 0.95^3 \\ &= 1 - 0.857 \\ &= 0.143 \end{aligned}$$

Five groups need  $m = 10$  tests:

$$1 - 0.95^{10} = 0.40$$



**Figure 2:** The chance of at least one false alarm: pairwise t-tests (red) against one ANOVA (blue), by formula and by simulation

In Figure 2, follow the red line: every extra group adds more pairs, and the risk climbs while ANOVA stays at 5%. So with five groups whose true means are all equal, pairwise t-tests “find” a difference far more often than 5% of the time. ANOVA asks one question about all groups at once, and its false-alarm rate stays at 5%.

**Extra:** The formula assumes the tests are independent. Pairwise t-tests are not, since each group appears in several of them, so the real risk is a little lower. The notebook measures the real risk on 2,000 simulated datasets (20 values per group, all true means equal):

| Groups | t-tests | At least one<br>false alarm<br>(simulated) | Formula | ANOVA false<br>alarm<br>(simulated) |
|--------|---------|--------------------------------------------|---------|-------------------------------------|
| 3      | 3       | 11.0%                                      | 14.3%   | 5.0%                                |
| 5      | 10      | 29.0%                                      | 40.1%   | 5.0%                                |

### 3. Splitting the variation

**Key point:** Total variation = variation within the groups + variation between the group means:  $SST = SSW + SSB$ , and the degrees of freedom split the same way.

**The data.** We carry one small dataset through the whole calculation: the marks (out of 10)

of 3 students in each of  $k = 3$  sections of a class.

| Section A | Section B | Section C |
|-----------|-----------|-----------|
| 4, 5, 6   | 6, 7, 8   | 8, 9, 10  |
| mean 5    | mean 7    | mean 9    |

The **grand mean** (G-866), the mean of all  $N = 9$  values, is:

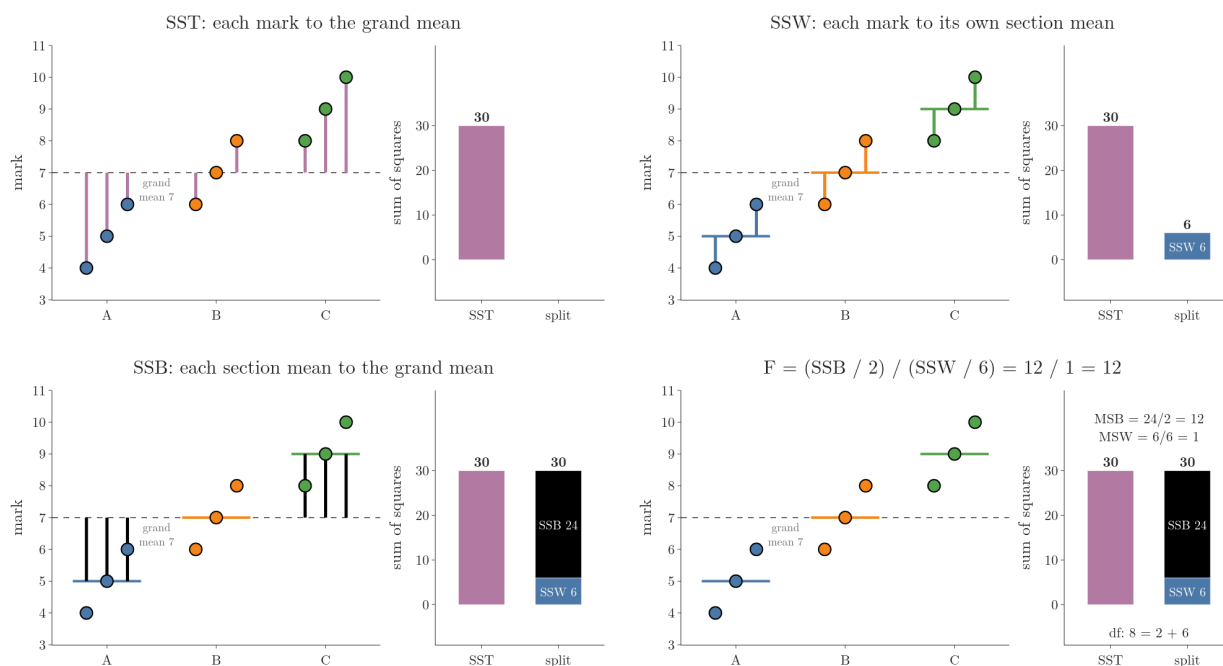
$$63/9 = 7$$

It is also the mean of the three section means, because the sections have equal sizes:

$$(5 + 7 + 9)/3 = 7$$

These are the left-hand groups of Figure 1.

**The idea in plain words.** The nine marks vary. Some of that variation is noise inside each section: students in the same section still score differently. The rest comes from the sections themselves having different means. ANOVA measures the two parts and compares them.



**Figure 3:** The nine marks as sticks: to the grand mean ( $SST = 30$ ), to their own section mean ( $SSW = 6$ ), and from each section mean to the grand mean ( $SSB = 24$ ); the bars stack  $30 = 24 + 6$ . Idea after Khan Academy, “ANOVA 2: Calculating SSW and SSB”

Variance is built from squared distances from a mean (see [variance](#)). ANOVA works with the **sums of squares** (G-1914), the squared distances added up before dividing. Figure 3 draws each distance as a stick; watch the sticks change as the reference line moves from the grand mean to the section means.

### 3.1 Total sum of squares

**Key point:**  $SST$  measures how far all values lie from the grand mean.

1. **In words:** square each value's distance from the grand mean and add over all values.
2. **Formula:**

$$SST = \sum_{\text{all values}} (x - \bar{x})^2$$

where  $\bar{x}$  is the grand mean and  $\sum$  means “add up” (here over all 9 values).

3. **Example:** distances from 7 are  $-3, -2, -1; -1, 0, 1; 1, 2, 3$ :

$$\begin{aligned} SST &= (9 + 4 + 1) + (1 + 0 + 1) \\ &\quad + (1 + 4 + 9) \\ SST &= 14 + 2 + 14 = 30 \end{aligned}$$

4. **Degrees of freedom:**  $N - 1 = 8$ . Once we know the grand mean and any 8 of the marks, the ninth is fixed, so only 8 marks carry free information.

### 3.2 Within-group sum of squares

**Key point:**  $SSW$  measures how far the values lie from their own group mean: the noise inside the groups.

1. **In words:** square each value's distance from its own group mean and add over all values.
2. **Formula:**

$$SSW = \sum_{\text{groups}} \sum_{\text{values in group}} (x - \bar{x}_i)^2$$

where  $\bar{x}_i$  is the mean of group  $i$ .

3. **Example:** each section gives:

$$1 + 0 + 1 = 2$$

So the three sections together give:

$$SSW = 2 + 2 + 2 = 6$$

4. **Degrees of freedom:**

$$N - k = 9 - 3 = 6$$

Within each section, once we know its mean and 2 of its 3 marks, the third is fixed: 2 free values per section, 3 sections. This is Bessel's correction (dividing by one less than the count) once per group (see [the sample variance](#)).

### 3.3 Between-group sum of squares

**Key point:**  $SSB$  measures how far the group means lie from the grand mean, each counted once per value in its group.

1. **In words:** replace every value by its group mean, then square that mean's distance from the grand mean and add over all values. Equivalently: square each group mean's distance from the grand mean, multiply by the group size, and add over groups.

2. **Formula:**

$$SSB = \sum_{\text{groups}} n_i(\bar{x}_i - \bar{x})^2$$

where  $n_i$  is the size of group  $i$ .

3. **Example:**

$$SSB = 3(5 - 7)^2 + 3(7 - 7)^2 + 3(9 - 7)^2$$

$$SSB = 12 + 0 + 12$$

$$SSB = 24$$

4. **Degrees of freedom:**  $k - 1 = 2$ . Once we know the grand mean and 2 of the 3 section means, the third is fixed.

### 3.4 The split

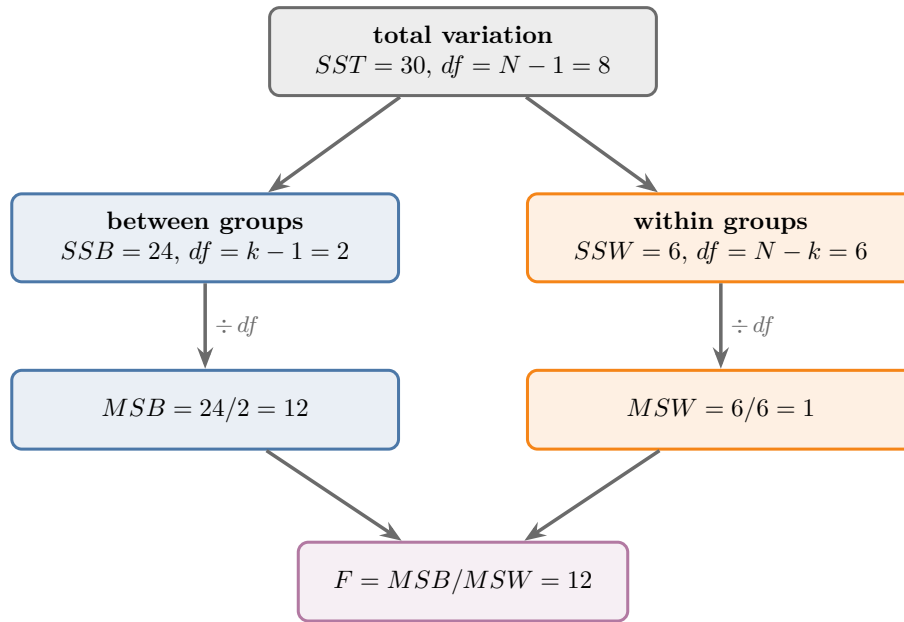
$$SST = SSB + SSW$$

$$30 = 24 + 6$$

The degrees of freedom split the same way:

$$8 = 2 + 6$$

Most of the variation here lies between the sections. Figure 4 summarises the split and the steps from there to F.



**Figure 4:** The variation split into its two parts, and the steps from there to F

## 4. The hypotheses

**Key point:**  $H_0$ : all group means are equal;  $H_1$ : at least one group mean differs from the others.

The sections' sample means differ: 5, 7 and 9. Is that a real difference between the sections, or just the luck of which students we happened to look at? The question is about the **population** means, the means over all students who could be taught in each section.

With  $k$  groups:

$$H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$$

$$H_1 : \text{at least one } \mu_i \text{ differs}$$

$H_1$  is not “all means differ”. Rejecting  $H_0$  tells us that some difference exists, not which groups differ; section 8 answers that. With three or more groups many patterns fit  $H_1$  (for example “A differs, B and C are equal”), so rejecting  $H_0$  points to “some difference”, not to one particular alternative.

“One-way” means one categorical feature defines the groups: here, the section.

## 5. The F statistic

**Key point:**  $F = MSB/MSW$ : the between-group variance divided by the within-group variance; a large  $F$  means the group means differ by more than the noise explains.

### 5.1 Mean squares

**Key point:** Dividing each sum of squares by its degrees of freedom turns it into a variance, a **mean square** (G-1200).

We use the degrees of freedom of section 3:  $k - 1 = 2$  between groups and  $N - k = 6$  within groups.

**The idea.** If  $H_0$  is true, the group means differ only through noise, so the between-group variance and the within-group variance measure the same noise and come out about equal. If the between-group variance is much larger, the variation comes mostly from differences between the means, and  $H_0$  looks wrong.

### 5.2 F

1. **In words:** divide each sum of squares by its degrees of freedom, then divide the between-group mean square by the within-group one.
2. **Formula:**

$$MSB = \frac{SSB}{k - 1}$$

$$MSW = \frac{SSW}{N - k}$$

$$F = \frac{MSB}{MSW}$$

3. **Example:**

$$MSB = \frac{24}{2} = 12$$

$$MSW = \frac{6}{6} = 1$$

$$F = \frac{12}{1} = 12$$

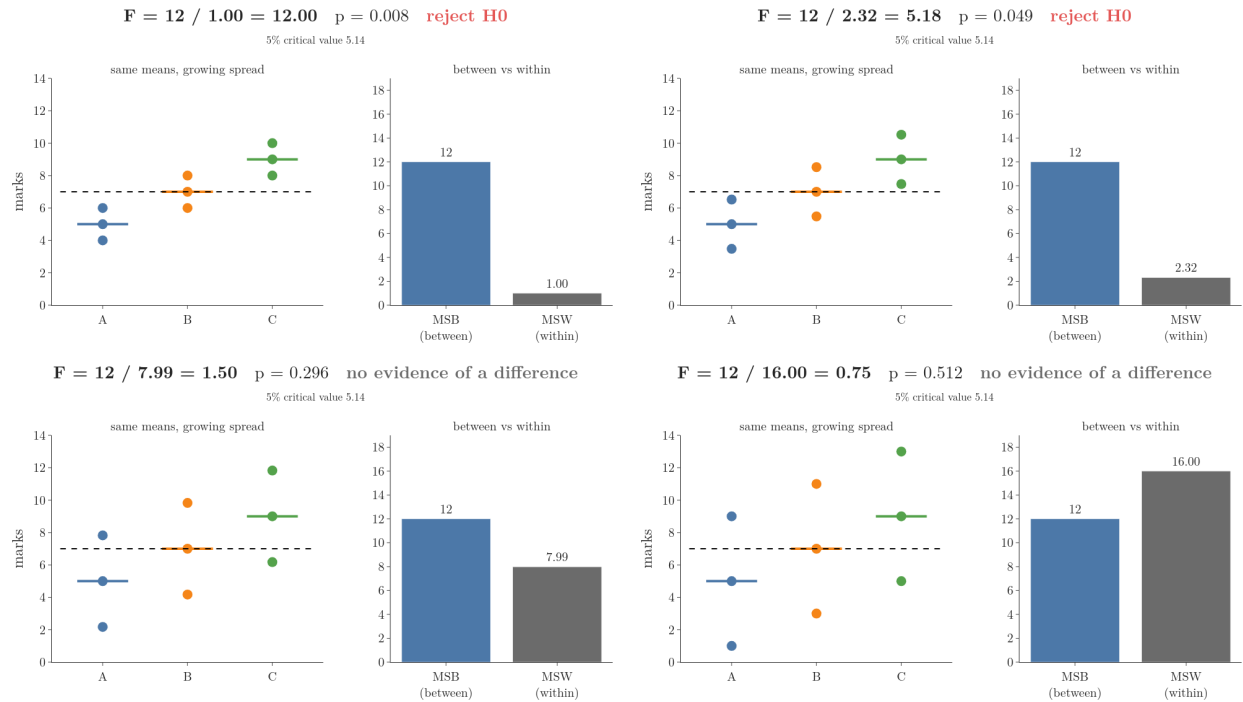
If  $H_0$  is true, both mean squares estimate the same noise variance, and  $F$  is close to 1 (Montgomery §3.3). Here the between-group variance is 12 times the within-group variance.

For the right-hand groups of Figure 1 (1, 5, 9; 3, 7, 11; 5, 9, 13) the means and so  $SSB = 24$  are the same, but  $SSW = 96$ . Then:

$$MSW = 96/6 = 16$$

$$F = 12/16 = 0.75$$

This is no evidence of a difference.



**Figure 5:** The groups of Figure 1 morphing from tight to spread out: MSB stays 12, MSW grows from 1 to 16, and  $F$  falls from 12 to 0.75

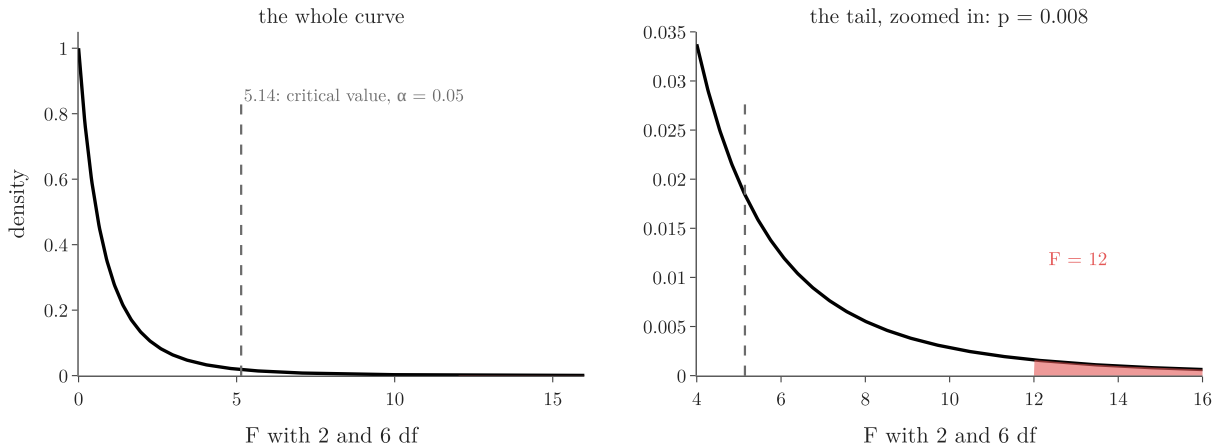
In Figure 5, watch the grey bar: only the within-group variance changes, and once  $F$  drops below the critical value 5.14 the same group means no longer count as different.

**Extra:** The results are usually set out as an **ANOVA table** (G-202):

| Source         | SS | df | MS | F  | p     |
|----------------|----|----|----|----|-------|
| between groups | 24 | 2  | 12 | 12 | 0.008 |
| within groups  | 6  | 6  | 1  |    |       |
| total          | 30 | 8  |    |    |       |

## 6. The F distribution and the decision

**Key point:** Under  $H_0$ ,  $F$  follows an F distribution with  $k - 1$  and  $N - k$  degrees of freedom; the p-value is its right-tail area beyond our  $F$ .



**Figure 6:** The F distribution with 2 and 6 degrees of freedom: the whole curve (left) and its tail (right)

The **F distribution** (G-739) is the distribution of a ratio of two independent variances (NIST Handbook §1.3.6.6.5). Like the chi-square distribution (see [the chi-square distribution](#)), it is never negative and is skewed to the right. The F distribution has two parameters: the degrees of freedom of the top and of the bottom variance.

Only a large  $F$  counts against  $H_0$  (group means further apart than noise explains), so the test is always right-tailed. Figure 6 shows the F distribution with 2 and 6 degrees of freedom:

- the 5% critical value is 5.14;
- our  $F = 12$  lies far beyond it;
- the tail area beyond 12 is  $p = 0.008$ .

The critical value depends on  $\alpha$ : at  $\alpha = 0.10$  it is 3.46, and  $F = 12$  is far beyond that too (notebook, section 2b).

Since  $0.008 \leq 0.05$ , we reject  $H_0$ : the three sections do not all have the same mean mark.

**Python:** `f_oneway` takes one array per group.

```
from scipy import stats
```

```
stats.f_oneway([4, 5, 6], [6, 7, 8], [8, 9, 10])
```

```
# statistic = 12.0, pvalue = 0.008
```

```
stats.f.ppf(0.95, 2, 6)      # 5.14, the critical value
```

**Extra:** With exactly two groups, ANOVA and the equal-variance two-sample t-test are the same test:  $F = t^2$  and the p-values match. For sections A and B alone,  $t^2 = 6$  and  $F = 6$ .

## 7. Assumptions

**Key point:** Independent observations, roughly normal values in each group, and equal variances across groups.

ANOVA makes the same assumptions as the two-sample t-test (see [its assumptions](#)), for every group:

1. **Independence.** The groups are separate (no one is in two groups), and the observations do not influence each other.
2. **Normality.** The values in each group are roughly normal. With large groups, the central limit theorem makes the group means close to normal anyway (see [the conditions of the central limit theorem](#), which give  $n \geq 30$  as the usual guide), so this matters less. Check with a Q-Q plot or the Shapiro-Wilk test (see [checking normality](#)).
3. **Equal variances.** The groups have about the same variance; Levene's test checks this, with  $H_0$ : "the variances are equal".

When the assumptions fail:

- unequal variances: **Welch's ANOVA** (G-2121) (Welch 1951), or `scipy's stats.alexandergovern`, drops the equal-variance assumption;
- strongly non-normal small groups: the **Kruskal-Wallis test** (G-1019) (`stats.kruskal`; Kruskal and Wallis 1952) compares the groups by ranks instead of means.

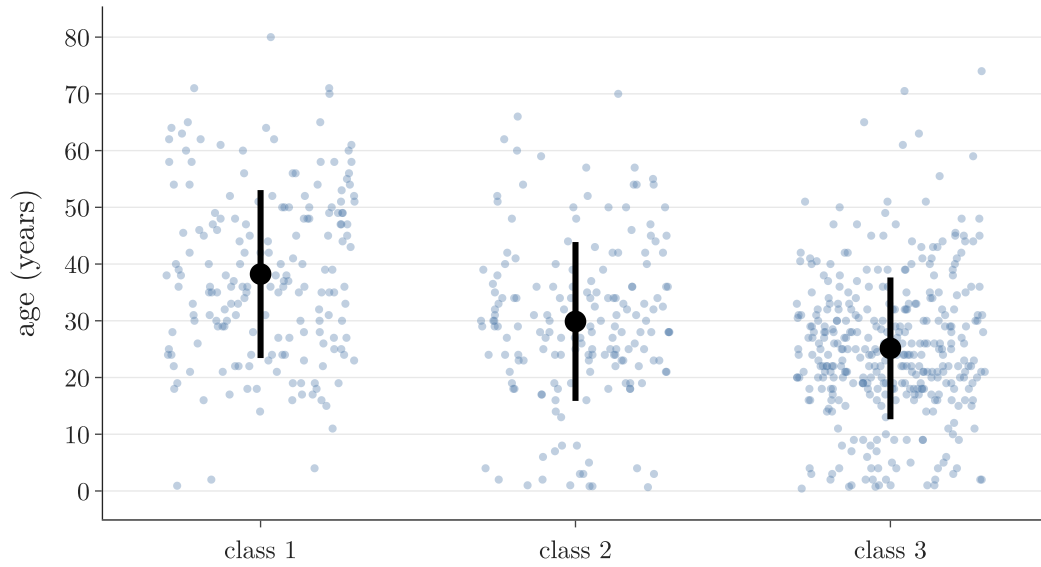
## 8. Case study: age by class on the Titanic

**Key point:** Mean ages of 38.2, 29.9 and 25.1 years in the three classes give  $F = 57.4$  and  $p \approx 10^{-23}$ ; Tukey's test shows that every pair of classes differs.

### 8.1 The test

**Key point:** Did the mean age of passengers differ between first, second and third class?

The Kaggle Titanic training file has 714 passengers with a known age.



**Figure 7:** Age of Titanic passengers by class: every passenger as a dot, the class mean and one standard deviation either side in black

| Class | Passengers | Mean age | Standard deviation |
|-------|------------|----------|--------------------|
| 1     | 186        | 38.2     | 14.8               |
| 2     | 173        | 29.9     | 14.0               |
| 3     | 355        | 25.1     | 12.5               |

Figure 7 shows a clear downward step in the means, with much overlap between the classes.

1. **Hypotheses.**  $H_0 : \mu_1 = \mu_2 = \mu_3$ ;  $H_1$ : at least one class mean differs.
2. **Significance level.**  $\alpha = 0.05$ .
3. **Statistic.** The two sums of squares are built as in the nine-mark example. *SSW* adds, over the classes, (passengers – 1) times the variance (the standard deviation squared). Roughly:

$$185 \times 14.8^2 = 40,522$$

$$172 \times 14.0^2 = 33,712$$

$$354 \times 12.5^2 = 55,313$$

$$SSW \approx 40,522 + 33,712 + 55,313$$

$$SSW \approx 129,547$$

*SSB* adds, over the classes, passengers times (class mean – overall mean 29.7) squared. Roughly:

$$186 \times 8.5^2 = 13,439$$

$$173 \times 0.2^2 = 7$$

$$355 \times 4.6^2 = 7,512$$

$$SSB \approx 13,439 + 7 + 7,512$$

$$SSB \approx 20,958$$

These table values are rounded. The unrounded data give  $SSB = 20,930$  with 2 df and  $SSW = 129,527$  with 711 df (the 714 passengers minus 3 classes):

$$MSB = \frac{20,930}{2} = 10,465$$

$$MSW = \frac{129,527}{711} = 182.2$$

$$F = \frac{10,465}{182.2} = 57.4$$

4. **P-value.** The p-value is far below 0.05; the 5% critical value is only 3.01:

$$p = 7.5 \times 10^{-24}$$

5. **Decide.** Reject  $H_0$ : mean age differed between the classes.

## 8.2 Checking the assumptions

**Key point:** Levene's test rejects equal variances ( $p = 0.004$ ), but the tests that drop that assumption agree: the classes differ.

- **Normality:** Shapiro-Wilk rejects normality for classes 2 and 3. With 173 and 355 passengers, the central limit theorem protects the means.
- **Equal variances:** Levene gives  $p = 0.004$ : the variances (standard deviations 14.8, 14.0, 12.5) differ.

Both checks raise doubts, so we confirm with the tests of section 7. Alexander-Govern gives a p-value of about  $4 \times 10^{-21}$  and Kruskal-Wallis about  $1 \times 10^{-21}$ . The conclusion stands.

## 8.3 Which classes differ? Tukey's test

**Key point:** A post-hoc test compares every pair of groups after ANOVA rejects, while keeping the overall false-alarm risk at 5%.

ANOVA said that at least one mean differs. A **post-hoc test** (G-1534) finds which. The usual choice is **Tukey's HSD** (G-2026) (honestly significant difference), which compares every pair with the overall Type I error held at  $\alpha$  (Tukey 1949):

| Pair              | Difference in mean age | p-value            |
|-------------------|------------------------|--------------------|
| class 1 – class 2 | 8.4 years              | $2 \times 10^{-8}$ |
| class 1 – class 3 | 13.1 years             | below $10^{-9}$    |
| class 2 – class 3 | 4.7 years              | 0.0005             |

| Pair | Difference in mean age | p-value |
|------|------------------------|---------|
|------|------------------------|---------|

All three pairs differ: first-class passengers were the oldest, third-class the youngest.

**Python:** ANOVA, the assumption checks and Tukey's test on the Titanic ages.

```
ages = [titanic.loc[titanic["Pclass"] == c, "Age"].dropna()
        for c in (1, 2, 3)]
stats.f_oneway(*ages)           # F = 57.4, p = 7.5e-24
stats.levene(*ages)            # p = 0.004
stats.alexandergovern(*ages)   # p = 4.4e-21
stats.tukey_hsd(*ages)         # every pair
```

## 9. ANOVA in machine learning

**Key point:** `f_classif` scores each numerical feature by its ANOVA F against the class labels, for feature selection.

`SelectKBest` (scores each column against the target and keeps the  $k$  best) can score features (the input variables of a model) with `f_classif` instead of `chi2`. For each numerical feature, `f_classif` computes the one-way ANOVA F with the classes of the **target** (G-1949) (the output we predict) as groups (scikit-learn docs, `f_classif`). A feature whose mean differs strongly between the classes gets a large F, and is kept.

## 10. Summary

|                       | Several t-tests             | One-way ANOVA                          |
|-----------------------|-----------------------------|----------------------------------------|
| Question              | is this pair different?     | do any of the $k$ means differ?        |
| Type I risk, 3 groups | about 14%                   | 5%                                     |
| Statistic             | $t$ per pair                | $F = MSB/MSW$                          |
| Distribution          | $t$ with $n_1 + n_2 - 2$ df | F with $k - 1$ and $N - k$ df          |
| Follow-up             | none needed                 | Tukey's HSD to find which pairs differ |

- $SST = SSB + SSW$ : total variation splits into variation between and within groups.
- $F$  compares the between-group variance with the within-group variance; near 1 under  $H_0$ , large when the means differ.
- The test is right-tailed; reject  $H_0$  when  $p \leq \alpha$ .
- Assumptions: independence, normality in each group, equal variances; Welch's ANOVA or Kruskal-Wallis when they fail.
- Rejecting  $H_0$  says some mean differs; Tukey's HSD says which.

## 11. Sources

### Built from

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## 12. Key terms

| Term                           | Meaning                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|
| One-way ANOVA                  | A test of whether three or more group means are equal, with the groups defined by one categorical feature       |
| Grand mean                     | The mean of all values from all groups together                                                                 |
| Sum of squares (SST, SSB, SSW) | Total, between-group and within-group squared distances; $SST = SSB + SSW$                                      |
| Mean square (MSB, MSW)         | A sum of squares divided by its degrees of freedom: a variance                                                  |
| F statistic                    | $MSB/MSW$ : between-group variance over within-group variance                                                   |
| F distribution                 | The distribution of a ratio of two variances; two degrees-of-freedom parameters, right-skewed                   |
| ANOVA table                    | The table of SS, df, MS, F and p for each source of variation                                                   |
| Familywise error rate          | The probability of at least one Type I error over several tests; $1 - (1 - \alpha)^m$ for $m$ independent tests |
| Post-hoc test                  | A test run after ANOVA rejects, to find which groups differ                                                     |
| Tukey’s HSD                    | A post-hoc test comparing every pair of groups with the overall Type I error held at $\alpha$                   |
| Welch’s ANOVA                  | A version of ANOVA that does not assume equal variances                                                         |
| Kruskal-Wallis test            | A rank-based alternative to one-way ANOVA that does not assume normality                                        |